

Global Semiconductors

Reality Check: Memory Upcycle Intact on Tight Inventories and Strong KV Cache Demand

CITI'S TAKE

Korea memory equities have seen share price pullback amid the debate around possible peak-out of DRAM/NAND cycle on weakened Chinese smartphone demand and rising NAND channel inventory. Contrary to the market concern, our market analysis indicates that NAND inventories stand extremely lean at 2.6 weeks at memory suppliers and 3 weeks at hyperscalers who are major eSSD customers. We reaffirm that solid AI demand keeps memory in undersupply. Reiterate Buy rating on Samsung and Hynix.

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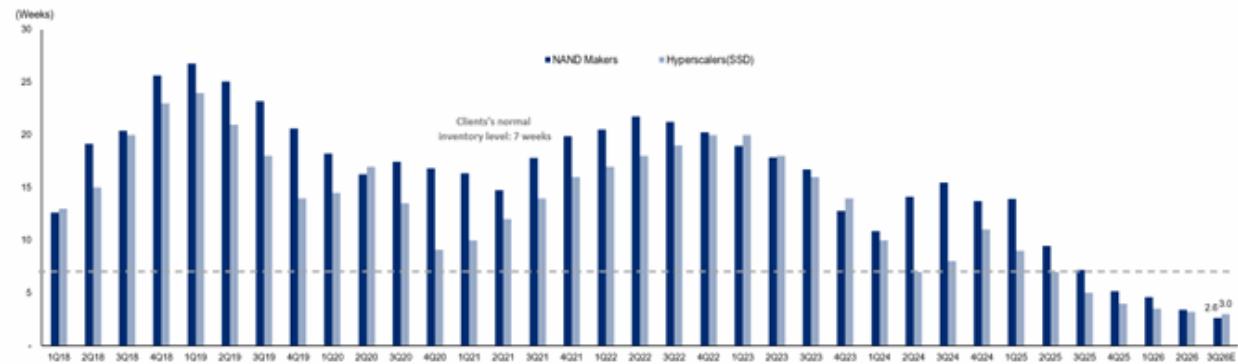
NAND Inventory of Memory Suppliers/Hyperscalers at 2.6/3.0 weeks — Our analysis indicates (Fig.1) that 3Q26 NAND inventory at NAND suppliers is severely depleted at 2.6 weeks, below the normal level of 5 weeks. Hyperscaler's NAND inventory is at 3 weeks, much lower than the normal level of 7 weeks. In terms of NAND channel inventory, the inventory is at 5 weeks, significantly lower than the average normal level of 15 weeks. Likewise, DRAM suppliers' inventory is at 2.7 weeks (Fig.2), lower than the normal level of 5 weeks. Hyperscalers' DRAM inventory remains very low at 2.5 weeks, falling below the normal level of 7 weeks. DRAM channel inventory is at 4 weeks, which is also exceptionally lower vs. the normal level of 15 weeks.

Robust NAND Demand Upside from CMX and QLC NAND despite Consumer NAND Weakness — While recent weakness in Chinese smartphone demand has raised concerns about softening NAND prices, we expect CMX demand to rise driven by KV cache demand growth on the increasing usage of AI agents. Moreover, we project QLC SSD NAND demand to grow further as QLC SSD NAND will be increasingly utilized as a near GPU storage solution to enhance AI computing efficiency. In particular, given that Nvidia's Vera Rubin has adopted 16TB TLC SSD for CMX operation, 1,152 TB SSD is required per Rubin server system. We project CMX NAND demand to reach 34.6/115.2bn 8Gb Equiv. in 26E/27E respectively, which translates to 2.8%/9.3% of 26E global NAND demand.

Implication — Contrary to market concerns, we find memory inventory levels to be materially low at both memory suppliers and their customers. Despite fears of weak Chinese smartphone demand, memory supply/demand sufficiency at DRAM/NAND suppliers has in fact fallen from 70% to 50%. With lean memory inventory situation across supply chain, rising CMX and QLC NAND demand will result in persistent undersupply. Reiterate Buy rating on Samsung and Hynix.

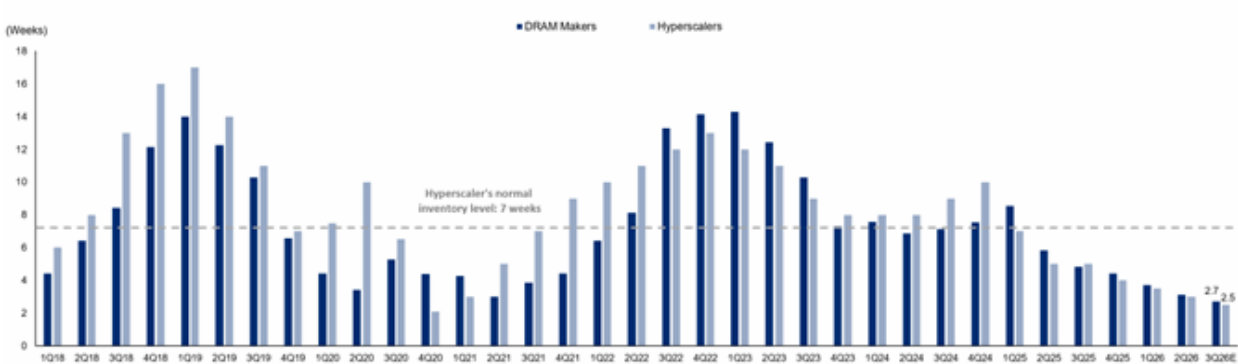
See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations

Figure 1. NAND Inventory Trend



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Source: Citi Research Estimates

Figure 2. DRAM Inventory Trend



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Source: Citi Research Estimates

Samsung Electronics

(005930.KS; W249500.0; 1; 24 Jul 26; 15:45)

Valuation

Our 12-month target price for Samsung of W530,000 is derived using a sum-of-the-parts (SOTP) methodology, based on 2026E EBITDA. In calculating total operating value, we reference global peers in assigning fair-value EV/EBITDA multiples for the five main divisions (7.6x for Memory, 4.1x for Foundry, 0.5x for Display Panel, 4.8x for Mobile and 2.0x for Consumer Electronics), in line with trading multiples of relevant peer companies.

Risks

Downside risks that could prevent the shares from reaching our target price include: 1) Longer-than-expected approval delay in HBM shipment to its key customers; 2) PC sales weaken more than our forecast and NAND demand fails to meet our expectations; 3) aggressive investment by competitors in memory semiconductor/foundry could have a negative impact on prices; 4) competition in the handset market intensifies, reducing SEC's handset margins; 5) any major appreciation of the won would impact SEC's earnings.

SK Hynix

(000660.KS; W1759000.0; 1; 24 Jul 26; 15:45)

Valuation

Our 12-month target price for SK Hynix of W3,100,000 is derived using a sum-of-the-parts (SOTP) methodology, based on 2026E EBITDA. In calculating total operating value, we break down SK Hynix's operating businesses into HBM and commodity/others business to reflect a structural change in next-gen memory market. We reference a global peer (TSMC) in assigning a fair-value EV/EBITDA multiple for HBM business, as we project the memory market is evolving from a traditional commodity market into a highly customized & customer-specific market much closer to the foundry business. For commodity memory business, we apply a historical average of 12m fwd EV/EBITDA during the beginning or early stage of memory upcycles in the past. We apply 7.2x EV/EBITDA to HBM segment and 4.2x to commodity segment to derive the target price.

Risks

Downside risks that could prevent the shares from reaching our target price include: 1) a downturn in DRAM demand; 2) weaker NAND demand than our forecasts; and 3) weakness in global consumption.

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Appendix A-1

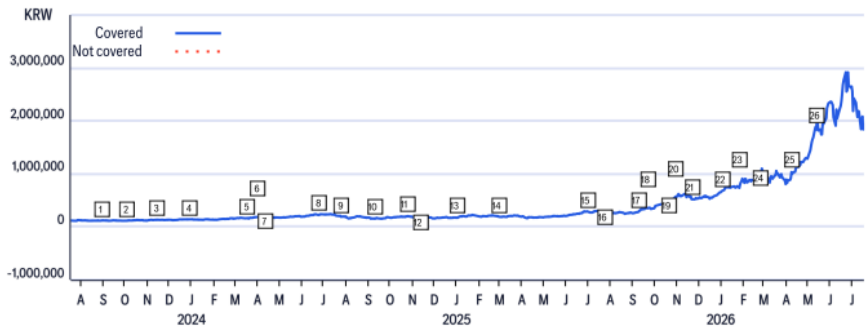
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IMPORTANT DISCLOSURES

SK Hynix (000660.KS)
Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



Date	Rating	Target Price	Closing Price
30-Aug-23 11:34:39	1	*180,000.00	119,400.00
04-Oct-23 06:14:16	1	*185,000.00	115,400.00
13-Nov-23 09:52:50	1	*190,000.00	131,800.00
01-Jan-24 18:20:58	1	*230,000.00	141,500.00
18-Mar-24 10:47:11	1	*234,000.00	164,300.00
01-Apr-24 10:34:31	1	*238,000.00	185,500.00
12-Apr-24 08:17:40	1	*310,000.00	187,400.00
25-Jun-24 06:27:18	1	*350,000.00	225,000.00
25-Jul-24 03:44:27	1	*337,000.00	190,000.00

Date	Rating	Target Price	Closing Price
09-Sep-24 02:31:32	1	*310,000.00	157,000.00
24-Oct-24 06:29:30	1	*330,000.00	198,200.00
12-Nov-24 10:31:16	1	*350,000.00	185,800.00
31-Dec-24 06:45:21	1	*340,000.00	173,900.00
03-Mar-25 15:00:11	1	*350,000.00	190,200.00
01-Jul-25 03:45:10	1	*430,000.00	285,500.00
24-Jul-25 04:40:40	1	*380,000.00	269,500.00
08-Sep-25 10:00:23	1	*430,000.00	277,000.00
22-Sep-25 06:19:28	1	*480,000.00	351,000.00

Date	Rating	Target Price	Closing Price
20-Oct-25 04:22:50	1	*640,000.00	485,500.00
29-Oct-25 02:38:54	1	*770,000.00	558,000.00
23-Nov-25 10:20:04	1	*830,000.00	521,000.00
02-Jan-26 03:22:42	1	*900,000.00	677,000.00
26-Jan-26 07:57:46	1	*1,400,000.00	736,000.00
24-Feb-26 08:08:02	1	*1,550,000.00	1,005,000.00
08-Apr-26 07:47:52	1	*1,700,000.00	1,033,000.00
11-May-26 07:42:14	1	*3,100,000.00	1,880,000.00

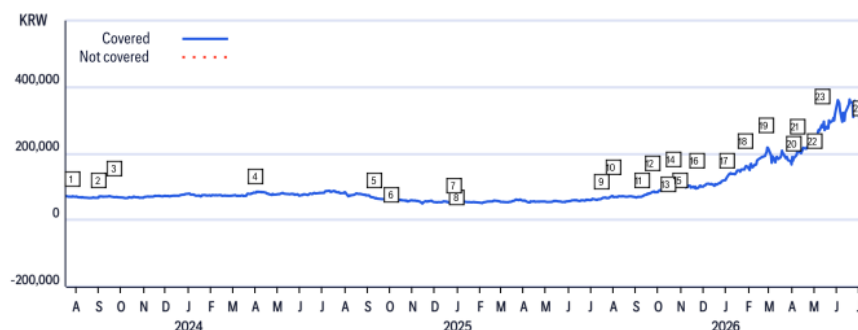
*Indicates Change

Rating/target price changes above reflect Eastern Time

Samsung Electronics (005930.KS)

Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



	Date	Rating	Target Price	Closing Price
1	27-Jul-23 05:53:47	1	*110,000.00	71,700.00
2	31-Aug-23 06:54:46	1	*120,000.00	66,900.00
3	22-Sep-23 09:53:22	1	*110,000.00	68,800.00
4	01-Apr-24 05:23:20	1	*120,000.00	82,000.00
5	09-Sep-24 02:31:35	1	*110,000.00	67,500.00
6	02-Oct-24 05:16:51	1	*97,000.00	61,300.00
7	26-Dec-24 07:42:50	1	*87,000.00	53,600.00
8	31-Dec-24 06:21:22	1	*83,000.00	53,200.00

*Indicates Change

	Date	Rating	Target Price	Closing Price
9	16-Jul-25 05:08:04	1	*90,000.00	64,700.00
10	31-Jul-25 02:49:57	1	*100,000.00	71,400.00
11	08-Sep-25 09:30:31	1	*110,000.00	70,100.00
12	22-Sep-25 05:57:03	1	*120,000.00	83,500.00
13	14-Oct-25 07:05:40	1	*133,000.00	91,600.00
14	20-Oct-25 06:25:10	1	*145,000.00	98,100.00
15	30-Oct-25 11:44:04	1	*150,000.00	104,100.00
16	23-Nov-25 10:07:18	1	*170,000.00	94,800.00

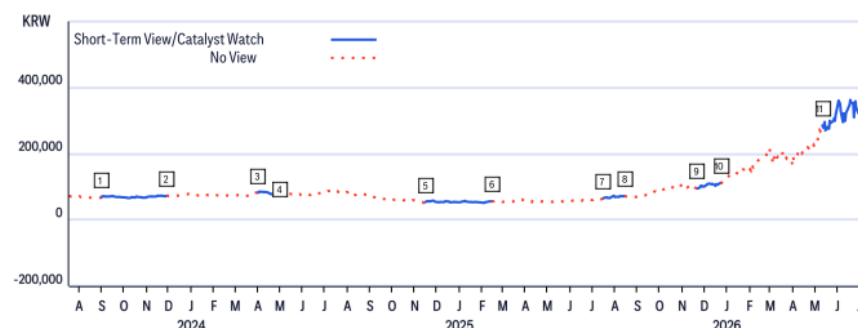
	Date	Rating	Target Price	Closing Price
17	02-Jan-26 02:41:08	1	*200,000.00	128,500.00
18	26-Jan-26 09:32:20	1	*240,000.00	152,100.00
19	24-Feb-26 05:11:17	1	*280,000.00	200,000.00
20	02-Apr-26 07:54:27	1	*300,000.00	178,400.00
21	07-Apr-26 02:31:16	1	*320,000.00	196,500.00
22	30-Apr-26 03:49:59	1	*300,000.00	220,500.00
23	11-May-26 07:42:55	1	*460,000.00	285,500.00
24	02-Jul-26 03:16:27	1	*530,000.00	286,000.00

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Samsung Electronics (005930.KS)

Short-Term View/Catalyst Watch Research

Analyst: Peter Lee



	Date	Action	Expected Direction	Duration	Closing Price
1	31-Aug-23 02:54:46	Add CW	Upside	90 Days	66,900.00
2	28-Nov-23 21:19:48	Remove CW	Upside	90 Days	72,700.00
3	01-Apr-24 01:23:20	Add CW	Upside	30 Days	82,000.00
4	30-Apr-24 22:52:33	Remove CW	Upside	30 Days	77,500.00

CW - Catalyst Watch, STV - Short-Term View

	Date	Action	Expected Direction	Duration	Closing Price
5	17-Nov-24 11:09:02	Add CW	Upside	90 Days	53,500.00
6	14-Feb-25 12:17:17	Remove CW	Upside	90 Days	56,000.00
7	16-Jul-25 01:08:04	Add CW	Upside	30 Days	64,700.00
8	15-Aug-25 14:07:20	Remove CW	Upside	30 Days	71,600.00

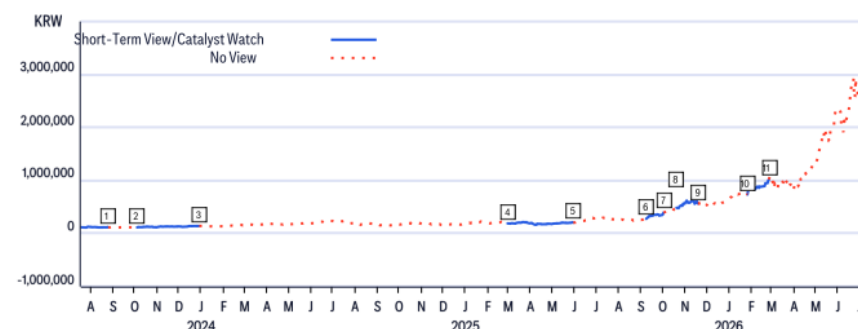
	Date	Action	Expected Direction	Duration	Closing Price
9	23-Nov-25 05:07:18	Add CW	Upside	90 Days	94,800.00
10	23-Dec-25 20:51:38	Remove CW	Upside	30 Days	111,500.00
11	11-May-26 03:42:55	Add CW	Upside	90 Days	285,500.00

Rating/target price changes above reflect Eastern Time

SK Hynix (000660.KS)

Short-Term View/Catalyst Watch Research

Analyst: Peter Lee



	Date	Action	Expected Direction	Duration	Closing Price
1	27-Aug-23 22:58:59	Remove CW	Upside	90 Days	116,500.00
2	04-Oct-23 02:14:16	Add CW	Upside	90 Days	115,400.00
3	01-Jan-24 21:19:41	Remove CW	Upside	90 Days	141,500.00
4	03-Mar-25 10:00:11	Add CW	Upside	90 Days	190,200.00

CW - Catalyst Watch, STV - Short-Term View

	Date	Action	Expected Direction	Duration	Closing Price
5	01-Jun-25 22:55:32	Remove CW	Upside	90 Days	204,500.00
6	08-Sep-25 06:00:23	Add CW	Upside	30 Days	277,000.00
7	08-Oct-25 22:50:00	Remove CW	Upside	30 Days	395,500.00
8	20-Oct-25 00:22:50	Add CW	Upside	30 Days	485,500.00

	Date	Action	Expected Direction	Duration	Closing Price
9	19-Nov-25 21:15:03	Remove CW	Upside	30 Days	562,000.00
10	26-Jan-26 02:57:46	Add CW	Upside	30 Days	736,000.00
11	25-Feb-26 20:57:35	Remove CW	Upside	30 Days	1,018,000.00

Rating/target price changes above reflect Eastern Time

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Citigroup Global Markets Korea Securities Limited

Peter Lee; Jayden Oh

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